

SURAKSHIT MATDAN

Surakshit Matdan — Secure Voting

*Quantum-Resistant, Zero-Knowledge,
Inclusive E-Voting Infrastructure for India*

Complete System Architecture, Integration Flow & Technical FAQ

India Innovates 2026 — Digital Democracy Domain — Secure E-Voting

IIT Jodhpur — M.Tech Artificial Intelligence — 2025–2027

System Philosophy & Scope

We are not replacing EVMs. We are extending democracy to the 30% of Indians EVMs cannot reach — and making the identity layer that feeds every election cryptographically airtight.

India’s electoral system has four structural gaps this system addresses:

60 Crore
Internal migrants cannot return to constituency on polling day. Their votes are permanently lost.

3.5 Crore
NRIs abroad with zero access to the Indian ballot despite being citizens.

~20 Million
Estimated fake/duplicate voter IDs in circulation, including Bangladeshi infiltrators enrolled via residence-only verification.

15 Years
Until quantum computers break RSA/ECC. Any e-voting system built without post-quantum crypto today needs full replacement by 2035–2040.

What This System Does — and Does Not — Do

THIS SYSTEM DOES	THIS SYSTEM DOES NOT
Digitize voter identity verification at booths	Replace EVMs or VVPAT infrastructure
Enable remote voting for NRI and migrants via ZKP	Guarantee zero surveillance (open source allows verification)
Prevent double voting mathematically via nullifiers	Solve the citizenship/NRC problem (it exposes it)
Future-proof against quantum cryptographic attacks	Work without national digital identity consensus
Merge EVM + remote tallies at ECI level	Store any voter identity post-credential generation

Three-Layer Architecture Overview

The system is designed as three independently deployable, air-gapped layers. Each layer can be adopted by the Election Commission of India separately.

Layer	Name	Function	Applies To	EVM Impact
Layer 1	Identity & Voter Roll	Aadhaar KYC + biometric verification + real-time voter roll digitization	All voters — booth + remote	Replaces paper lists only
Layer 2	ZKP Vote Casting	Zero-knowledge proof + post-quantum encryption + blockchain submission	Remote voters only (NRI, migrants)	Zero — runs in parallel
Layer 3	Counting & Tally	Homomorphic decryption + EVM merge + unified ECI result	Both channels	Adds remote tally only

Air-Gap Design Principle: Layer 1 and Layer 2 never communicate directly after credential handoff. Layer 1 knows *who you are* but not how you voted. Layer 2 knows *a valid credential voted* but not whose. Layer 3 knows aggregate counts but no individual votes. No single layer compromise can link identity to vote choice.

Layer 1 — Identity & Voter Roll Digitization

Layer 1 answers one question with cryptographic certainty: Is this person a real, eligible, unique Indian voter who has not already voted today?

3.1 Aadhaar Integration — Present & Future

Phase 1 (Now): Offline KYC

Voter downloads UIDAI-signed XML from Digilocker or UIDAI portal themselves.

- No Aadhaar number in XML
- UIDAI never knows where used
- Legally compliant with Aadhaar Act
- No API dependency during election

Phase 2 (Future): Aadhaar Online API

When Aadhaar becomes India's single national identity, direct API enables real-time biometric authentication.

- UIDAI Auth API v2.5
- Fingerprint + iris + OTP modes
- Requires AUA/KUA licensing from UIDAI
- Needs NRC foundation first (Section 6)

3.2 Complete Layer 1 Flow

01 Voter Arrives

Physical booth: approaches booth worker tablet. Remote voter: opens web app in browser. Feature phone user: approaches government Assisted Voting Booth running same system on a tablet.

02 Aadhaar KYC Submission

Phase 1: Voter uploads Offline KYC XML. System verifies UIDAI RSA-2048 digital signature locally — no server call. Extracts: name, age, address, photo. Confirms age 18+, maps constituency. XML deleted immediately after extraction.

03 Passive Liveness Detection

MediaPipe FaceMesh analyzes single frame for texture depth, specular reflection, micro-expression patterns. Runs entirely in WebAssembly on-device. Blocks: printed photos, screen replays, static masks. Silent — voter takes no action.

04 Active Liveness Challenge

System randomly selects one of 5 prompts: blink / turn left / smile / look up / nod. Voter performs action. Randomness blocks pre-recorded video replay and deepfake preparation.

05 Face Biometric Match

Live face compared against KYC XML photo using FaceNet/ArcFace (ONNX, WASM, on-device). Threshold: 98.5% cosine similarity. On match: SHA-3 hash of embedding generated; raw biometric immediately discarded. PDPB compliant.

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Deduplication Check

Face hash queried against national biometric deduplication registry. Ensures same biological person not enrolled twice under different voter IDs. Flags duplicates for ECI review — human review required for edge cases.
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Voter Roll Sync

Voter marked *appeared* in real-time digital voter roll on NIC Cloud. Syncs across all booths in constituency via WebSocket. Booth worker sees: GREEN (verified), RED (already appeared elsewhere), AMBER (biometric mismatch). Offline fallback with queue.
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Voting Credential Generation

CRYSTALS-Kyber key pair generated. Private key (voting credential) sent only to voter’s device — never stored on server. Election-day expiry. All identity data purged from server at this point.

3.3 Layer 1 Technology Stack

Component		Technology	Why This Choice
Identity document		Aadhaar Offline XML / UIDAI Auth API v2.5	Only legally compliant path under Aadhaar Act. Offline avoids API dependency during election.
Signature verification		RSA-2048 (UIDAI public key)	UIDAI signs all Offline KYC XMLs. Local verification, no network needed.
Liveness detection		MediaPipe + Silent Face Spoofing	Runs in WebAssembly. Open source. No biometric leaves device.
Face matching		FaceNet / ArcFace (ONNX, WASM)	State-of-art accuracy. On-device inference, no cloud API.
Face hashing		SHA-3 256-bit	One-way, quantum-resistant. Face unrecoverable from hash.
Credential crypto		CRYSTALS-Kyber (liboqs WASM)	NIST PQC standard 2024. Quantum-resistant key encapsulation.
Voter database	roll	PostgreSQL + Redis (NIC Cloud)	PostgreSQL for durability. Redis for real-time booth sync (sub-100ms).
Multilingual UI		Bhashini API	22 Indian languages. Government-built. Free.
Hosting		NIC Cloud	India-sovereign. Data never leaves Indian jurisdiction.

Layer 2 — ZKP Vote Casting (Primary Build)

Layer 2 is what we are building for this hackathon. It is a self-contained, browser-based cryptographic voting engine. No government infrastructure dependency. Fully demoable as a standalone web application.

4.1 Zero-Knowledge Proof — Plain English

A Zero-Knowledge Proof lets you prove a statement is true without revealing *any* information about why it is true.

Analogy: Proving you know the combination to a safe — without opening it, without saying the numbers, without giving any clue — just by convincing a mathematician that you know. The mathematician becomes certain you know the combination but learns nothing about what it is.

In our system, the ZKP simultaneously proves three things without revealing any of them:

- I possess a valid voting credential issued for this specific election
- I have not cast a vote in this election before (my nullifier is fresh)
- My vote is for exactly one candidate — not blank, not multiple

The blockchain never sees who you are. It receives only a mathematical proof that all three conditions are satisfied by someone entitled to vote. It cannot determine whose proof it is.

4.2 The Nullifier — How Double-Voting is Prevented

Generation

Nullifier =
SHA-3(Credential +
Election_ID)

Same person, same election
= always identical nullifier.
Different person or election
= completely different.

Publication

When vote is cast, nullifier
is published on blockchain
publicly.

Anyone can see used nullifiers.
Nobody can reverse a
nullifier back to identity.

Enforcement

Second vote attempt generates
identical nullifier.

Blockchain sees it exists. Rejects.
Mathematical rejection — no human needed.

4.3 Complete Layer 2 Flow

01 Credential Received from Layer 1

Voting credential arrives in browser memory from Layer 1 handoff. Layer 2 has no knowledge of who the credential belongs to. Valid only for current election day.

02 Nullifier Check

Browser computes nullifier locally: SHA-3(credential + election_id). Queries blockchain nullifier registry. If found: REJECTED — mathematical double-vote prevention. If not found: voter proceeds.

03 Candidate Selection

Voter selects candidate from constituency list. UI available in 22 Indian languages via Bhashini. Accessible design for differently-abled voters.

04 ZKP Circuit Execution (On-Device)

snarkjs (Groth16 proving system) runs in WebAssembly inside browser. Circuit proves: valid credential + fresh nullifier + single candidate. Proof generation: 2–5 seconds on modern smartphone. No vote data sent to server during this step.

05 Post-Quantum Vote Encryption

Vote encrypted using CRYSTALS-Dilithium (NIST PQC 2024) with election authority's public key. Runs in liboqs WASM in browser. Encrypted vote is meaningless without 2-of-3 threshold decryption key held jointly by ECI + Supreme Court + independent auditor.

06 Blockchain Submission

Three items sent to Hyperledger Fabric: (1) Encrypted vote, (2) ZKP proof, (3) Nullifier. Each node independently verifies ZKP proof before accepting. 3-of-5 consensus required. Nodes: ECI + Supreme Court + 3 independent national auditors.

07 Nullifier Publication & Receipt

Nullifier published publicly. Voter receives anonymous receipt: SHA-3 hash of encrypted vote. Proves vote was recorded. Cannot reveal vote choice.

4.4 Layer 2 Technology Stack

Component	Technology		Detail
Web platform	Progressive Web	App	No app store. Inspectable in browser DevTools. Works on any device.
ZKP system	snarkjs	+ circom	Runs in WebAssembly. Proof generation entirely on voter's device.
Post-quantum signing	CRYSTALS-Dilithium (liboqs WASM)		NIST PQC standard 2024. Quantum-resistant for 50+ years.
Post-quantum KEM	CRYSTALS-Kyber (liboqs WASM)		NIST PQC standard 2024. Key encapsulation for vote encryption.
Nullifier hashing	SHA-3	256-bit (native browser)	Quantum-resistant. Built into all modern browsers natively.
Vote counting	Homomorphic encryption	(Microsoft SEAL)	Votes tallied without individual decryption. Mathematical privacy.
Blockchain	Hyperledger Fabric 2.5		Permissioned. Enterprise-grade. No gas fees. 3-of-5 consensus.
Threshold decryption	Shamir Secret	Sharing (2-of-3)	Election key split across ECI + Supreme Court + auditor.
Backend API	Node.js + Express		Stateless. Receives only encrypted vote + proof + nullifier. No identity.
Multilingual	Bhashini API		22 Indian languages. Government-built. Free tier.

Layer 3 — Vote Counting & Unified Tally

5.1 Stream A — Physical EVM Count (Unchanged)

- Physical booth voters cast votes on EVM hardware as per current ECI protocol — zero change
- VVPAT paper trail remains as existing safeguard
- EVM count completed at counting centre by Returning Officer
- Result: constituency-wise EVM vote tally

5.2 Stream B — Blockchain Remote Vote Count

01 Poll Close — Threshold Decryption

ECI + Supreme Court + auditor jointly execute 2-of-3 Shamir Secret Sharing decryption. No single entity can trigger decryption alone. Fully logged and publicly auditable.

02 ZKP Proof Batch Verification

Each of 5 blockchain nodes independently verifies every ZKP proof. Invalid proofs flagged and excluded. 1 crore votes verifiable in under 2 hours (fully parallelizable).

03 Homomorphic Tally

Microsoft SEAL homomorphic encryption allows votes to be summed without decrypting individually. Result is constituency-wise count. No individual vote ever revealed — not even to ECI.

04 Public Audit Publication

Complete nullifier list published. Complete ZKP proof list published. Any citizen or journalist can independently verify: total count is correct, no vote counted twice, every counted vote had a valid proof.

5.3 Unified Result

Final Result = EVM Tally (Stream A) + Blockchain Remote Tally (Stream B)

Merged at constituency level by Returning Officer. Both streams independently auditable. Discrepancy protocol: if Stream B count exceeds registered remote voter list, full audit triggered automatically.

The Foundational Identity Problem — Fake Voters & NRC

Honest assessment: This system cannot solve India's fake voter problem alone. But it makes the problem visible, auditable, and fixable for the first time.

6.1 Why Fake Voters Exist Today

- Voter ID enrollment is residence-based — not citizenship-based
- Aadhaar proves residence, not citizenship — ~12 crore Aadhaar issued to non-citizens by UIDAI's own estimates
- No single database links birth records + parentage + continuous residence + biometric uniqueness
- Bangladeshi infiltrators enrolled in West Bengal, Assam, Delhi through normal registration with fraudulent documents
- Estimated 20 million fake/duplicate voter IDs in current electoral rolls

6.2 What This System Does About It

What it solves:

- Biometric dedup catches same person enrolled twice
- Face match prevents proxy voting
- ZKP nullifier prevents double voting
- Digitized roll enables post-election audit

What it does NOT solve:

- Bangladeshi with forged docs still passes Layer 1
- Citizenship verification requires NRC — political decision
- Aadhaar itself has ~12 crore non-citizen enrollments
- Governance problem, not a cryptography problem

6.3 The NRC Requirement

For Layer 1 to achieve true citizenship-level verification, India requires a National Register of Citizens (NRC) linked to birth records, parentage, and continuous residence, integrated with Aadhaar biometrics for a single source of truth. This is a political and administrative decision — not a technical one.

We are building the infrastructure that makes NRC-grade voter verification technically possible. Whether India implements NRC is a democratic decision beyond the scope of this system. Until then, this system significantly raises the cost of fake voter infiltration — it does not eliminate it.

Complete Security & Attack Matrix

Attack Vector	Technical Defence	Layer	Residual Risk
Fake Aadhaar document	UIDAI RSA-2048 signature verified locally	Layer 1	Near zero
Wrong person using valid ID	Face biometric match at 98.5% threshold	Layer 1	Low
Photo / screen replay	Passive liveness (texture/depth analysis)	Layer 1	Low-Med
Pre-recorded video replay	Active liveness (random prompt)	Layer 1	Very low
Deepfake live video	Passive + active combined liveness	Layer 1	Medium
Voting twice	ZKP nullifier — deterministic, blockchain-published	Layer 2	Zero
Intercepting vote in transit	CRYSTALS-Dilithium encryption before transmission	Layer 2	Zero
Quantum computer (2035+)	CRYSTALS-Kyber + Dilithium (NIST PQC 2024)	Layer 2	Zero (50yr)
Server breach for identities	No identity stored post-credential generation	All	Zero
Coerced voting	No linkable receipt — ZKP non-attributable	Layer 2	Low
Blockchain node compromise	3-of-5 consensus — majority requires 3 nodes	Layer 2	Low
Malicious JS served by server	Published SHA-3 hash of JS bundle — verifiable	Layer 2	Low
Fake voter with forged docs	Biometric dedup; citizenship gap remains	Layer 1	Medium
Tally manipulation	All proofs public — independent verification	Layer 3	Zero

Technical FAQ — Privacy, Security & Architecture

Q: How will citizens trust they are not being tracked?

They should not trust — they should **verify**. The entire system is open source, published on GitHub. Any citizen, journalist, or cryptographer can read every line of code and confirm no identity is transmitted. Additionally, reproducible builds allow anyone to compile the source and verify the deployed application is byte-for-byte identical to what is published. Trust requires faith in institutions. We offer mathematical verifiability instead.

Q: Can someone create a fake proof?

A fake ZKP proof is computationally infeasible without a valid voting credential. The ZKP circuit requires knowledge of the credential as input — without it, the prover cannot satisfy the circuit constraints. Even with a credential, you cannot generate a fresh nullifier if you have already voted — the nullifier is deterministic and already published on the blockchain. The only path is to defeat all of Layer 1 first.

Q: How does ZKP ensure the same person cannot vote twice?

The nullifier is a one-way hash of (Voting Credential + Election ID). Same person in same election always produces the same nullifier. When the first vote is cast, the nullifier is published publicly on blockchain. Any subsequent attempt generates the identical nullifier. The blockchain rejects it. Enforced by mathematics, not by any human or institution.

Q: What is a one-way function? Can quantum computers reverse it?

A one-way function is easy to compute forward and computationally infeasible to reverse — like mixing paint (easy) versus unmixing (impossible). In our system: $\text{SHA-3}(\text{credential} + \text{election_id}) = \text{nullifier}$. Going backward is infeasible. Against quantum computers: SHA-3 is specifically quantum-resistant. Grover's algorithm only halves effective security — SHA-3 256-bit becomes effectively 128-bit secure, which remains computationally infeasible for any foreseeable quantum hardware.

Q: Can we generate quantum-secured proofs in a browser?

Yes. WebAssembly (WASM) runs near-native speed inside any modern browser. Both snarkjs (ZKP) and liboqs (post-quantum crypto) compile to WASM. The entire cryptographic stack runs inside the browser tab on the voter's own device. ZKP proof generation takes 2–5 seconds on a modern smartphone. CRYSTALS-Dilithium signing takes under 1 second. Nothing cryptographically sensitive touches the network — only the final encrypted output leaves the browser.

Q: Why a web app instead of a native app?

Web is stronger for trust. No app store means Apple/Google cannot pull or modify the app silently. No installation means nothing persists after the session. Source code is inspectable live in browser DevTools by

any user. Works on any device with a browser. The only weakness: HTTPS requires trusting the server to serve correct code. Mitigated by publishing a cryptographic hash of the correct JavaScript bundle that any user can verify.

Q: What about people with keypad/feature phones?

The web app runs on any smartphone browser. For feature phone users, government-operated Assisted Voting Booths run the same system on a tablet. The booth is simply a hardware client — same Layer 2 stack, same cryptographic guarantees, same blockchain, same privacy. No separate backend, no reduced security. Every Indian votes with identical cryptographic protection regardless of device.

Q: Why will the government adopt this instead of keeping EVMs?

We are not asking them to. EVMs have Rs.11,000 crore in infrastructure and 25 years of operational trust. This targets three gaps EVMs structurally cannot solve: (1) NRI voting — 3.5 crore citizens with no current solution, a 10-year ECI priority. (2) Migrant worker voting — 60 crore permanently disenfranchised. (3) Panchayat elections — 250,000 gram panchayats on paper ballots with rampant fraud. Pilot path: panchayat elections first, then state elections for remote voters. Not revolution — expansion.

Q: How does Aadhaar Offline KYC differ from Aadhaar API and why is it legally safer?

Aadhaar API requires UIDAI authentication — it logs who authenticated, when, and from where. The Aadhaar Act restricts authentication to UIDAI-approved use cases. Offline KYC is different: voter downloads a UIDAI-digitally-signed XML themselves. We verify UIDAI's signature locally — no API call to UIDAI during verification. UIDAI never knows the XML was used for voting. The XML contains no Aadhaar number. Legally compliant because we are verifying a government-signed document, not authenticating via Aadhaar.

Q: Can the government track how I voted even with ZKP?

No, by mathematical design. The government receives: an encrypted vote (meaningless without 2-of-3 threshold key), a ZKP proof (proves validity, reveals nothing about content), and a nullifier (one-way hash, irreversible). Even if ECI, the Supreme Court, and an independent auditor all colluded — they could decrypt the aggregate tally via homomorphic counting but could not link any individual encrypted vote to any individual voter. The linkage was severed in Layer 1 before the vote was ever cast.

One Vote. One Identity. Unbreakable.

For every Indian who has ever been too far from home to vote.

And for every voice that was silenced.
